ORBITAL

Operational Reusable Backend for Integrated Trajectory and Logistics

Technical Report

Abstract

ORBITAL is a modular, domain-agnostic mission planning framework designed for constraint-critical aerospace systems. Modern aerospace operations require planning architectures capable of balancing performance objectives against strict operational constraints under uncertainty. Traditional planners are often tightly coupled to specific vehicle domains, limiting reuse and increasing engineering complexity.

ORBITAL separates optimization mechanics from domain-specific physics. The framework implements a reusable optimization kernel capable of generating executable mission plans across heterogeneous domains—including fixed-wing UAV trajectory planning and LEO CubeSat scheduling—without modification to the core planner.

This report presents the mathematical formulation, architecture, optimization strategy, robustness framework, validation results, and development roadmap of ORBITAL.

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1. Problem Formulation

Aerospace mission planning is formalized as a constrained optimization problem.

Let:

- $\mathbf{x} \in \mathbb{R}^n$ represent decision variables
- $\mathbf{J(x)}$ represent objective cost
- $\mathbf{C_i(x)}$ represent constraint margins

We solve:

Minimize

$J(x)$

Subject to

$C_i(x) \geq 0$ for all hard constraints

Constraint margins quantify distance from violation, enabling continuous visibility into feasibility.

A foundational principle of ORBITAL is:

> Feasibility precedes optimality.

Constraint enforcement is separated from objective ranking to preserve safety discipline.

1.1 Decision Variables

Decision variables encode controllable mission parameters, including:

- Velocity profiles
- Waypoint ordering (permutation variables)
- Timing schedules
- Resource allocation
- Observation sequencing
- Control intensities

Supported variable types:

- Continuous
- Integer
- Discrete

- Permutation

The resulting search space is high-dimensional, non-convex, and combinatorial. The optimization engine makes no convexity or differentiability assumptions.

1.2 Constraints

Constraints are evaluated using margin-based feasibility:

- Margin ≥ 0 → feasible
- Margin < 0 → violation

Hard Constraints

Hard constraints represent safety-critical requirements:

- Battery state ≥ 0
- Fuel ≥ 0
- Bank angle within limits
- Slew rate limits
- Geofence boundaries
- Ground-station visibility thresholds

Any hard violation renders the solution infeasible.

Soft Constraints

Soft constraints represent performance preferences and incur penalties but do not invalidate feasibility.

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2. Objective and Penalty Formulation

All objectives are expressed in cost space. Maximization problems are transformed via sign inversion.

The total score is:

$$\mathbf{S = J(x) + w \cdot \Sigma\ (max(0, -C_i(x)))^2}$$

Where:

- $J(x)$ = objective cost
- w = penalty weight
- $\max(0, -C_i(x))$ = violation magnitude

Squaring violations:

- Amplifies severe breaches
- Maintains smooth penalty scaling
- Encourages shallow violations to resolve

Hard constraint feasibility is evaluated independently of score ranking.

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3. System Architecture

ORBITAL consists of four decoupled layers:

1. Decision Variables — search space representation
2. Simulation — domain-specific physics
3. Constraints — margin-based feasibility
4. Objective — performance scoring

The optimization kernel operates independently of simulation logic. This enables cross-domain reuse without modifying the planner.

Key architectural properties:

- Modularity
- Extensibility
- Deterministic evaluation
- Cross-domain scalability

3.1 Execution Lifecycle

For each candidate solution:

1. Sample or mutate decision variables
2. Construct executable plan
3. Run domain simulation
4. Evaluate constraint margins
5. Compute objective
6. Aggregate penalties
7. Rank solution
8. Iterate

This structured lifecycle preserves clarity and reproducibility.

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4. Domain Physics Integration

4.1 Aircraft Domain

The UAV domain includes:

- Wind-aware kinematics
- Bank-angle-constrained turn radius
- Energy tracking
- Geofence enforcement

Results:

- 27.6% fuel reduction vs baseline routing
- Zero hard constraint violations

4.2 Spacecraft Domain

The spacecraft domain integrates physics-based orbital modeling:

- Two-body propagation
- J2 perturbation
- RK4 numerical integration
- Inertial state propagation (ECI)
- ECI → ECEF transformation via GMST (IAU 1982)
- Velocity frame correction ($\omega \times r$)
- Elevation-angle-based ground-station visibility

Separating inertial propagation from Earth-fixed operational evaluation ensures correct access timing and ground track generation.

Seven-day LEO simulation achieved:

- 87.5% target coverage
- 93.8% data return rate
- Zero hard constraint violations

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5. Optimization Strategy

ORBITAL employs mutation-based heuristic search with annealed mutation intensity.

Characteristics:

- Gradient-free
- Handles discrete and combinatorial variables
- Suitable for non-convex problems

Mutation amplitude decreases over iterations, transitioning from exploration to exploitation.

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6. Robustness Under Uncertainty

ORBITAL incorporates multi-scenario robustness evaluation.

Uncertainty modeling includes:

- Wind ±30%
- Mass ±5%
- Drag ±10%

Monte Carlo evaluation across 100+ seeds yielded:

- 97% hard-constraint feasibility rate
- Deterministic reproducibility via controlled RNG seeding

Risk-aware scoring supports CVaR-style aggregation beyond nominal performance.

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7. Validation Results

Two demonstration domains validated cross-domain capability.

UAV Multi-Waypoint Mission

- 27.6% fuel reduction
- Zero hard violations

CubeSat 7-Day Observation Mission

- 87.5% coverage
- 93.8% data return
- Deterministic execution
- Frame-consistent access modeling

The unified optimization kernel required no modification between domains.

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8. Limitations

Current limitations:

- Heuristic search without deterministic optimality guarantees

- Limited benchmarking against large-scale constellations
- Simplified drag and environmental modeling
- No distributed evaluation framework

The system is suitable for research and prototype validation but not yet flight-certified.

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9. Future Development

Planned enhancements:

- Parallel scenario evaluation
- Constellation-level coordination
- Adaptive penalty weighting
- Higher-fidelity environmental models
- Enhanced margin visualization dashboards

Long-term objective: evolve ORBITAL into a scalable optimization backend supporting high-density aerospace infrastructure.

Conclusion

ORBITAL demonstrates that a reusable, domain-agnostic optimization kernel can generate physically executable mission plans across heterogeneous aerospace domains while preserving strict feasibility discipline.

By decoupling decision variables, simulation physics, constraint evaluation, and objective scoring, ORBITAL establishes a scalable architecture for structured decision-making under constraints—an essential capability in modern aerospace systems engineering.